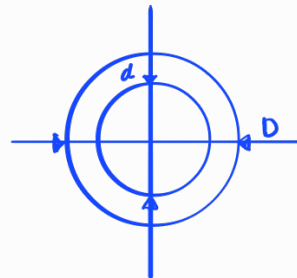
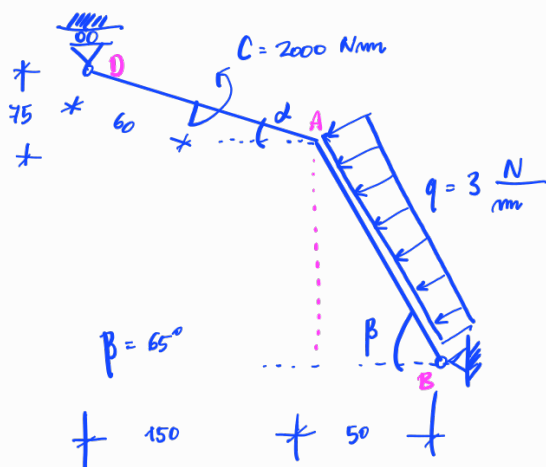


Calcolo dello spostamento



$$d = 6 \text{ mm}$$

$$D = 10 \text{ mm}$$

Dato la struttura in acciaio riportata in figura, si richiede di:

- calcolare lo spostamento orizzontale del punto A;
- tracciare i diagrammi delle azioni interne;
- calcolare lo sforzo massimo normale della struttura;

$$\overline{AD} = \sqrt{75^2 + 150^2} = 167.7 \text{ mm}$$

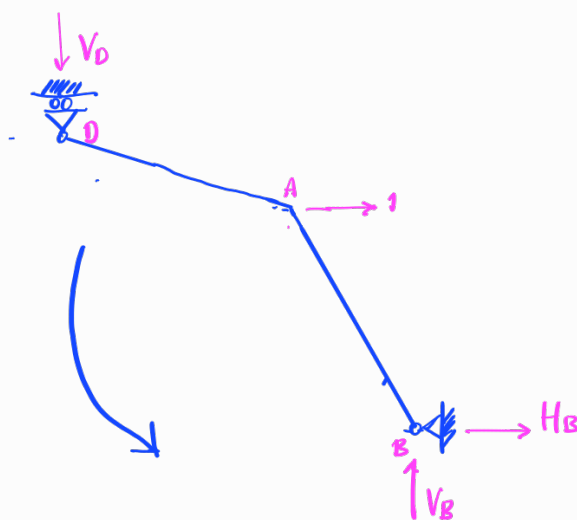
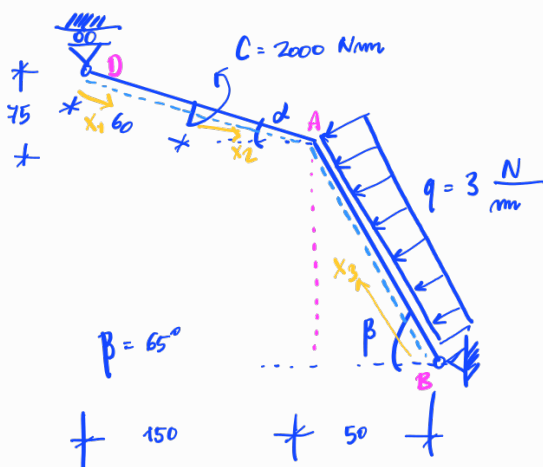
$$\overline{AB} = \frac{50}{\cos \beta} = 118.31 \text{ mm}$$

$$\alpha = \arctan\left(\frac{75}{150}\right) = 26.56^\circ$$

$$A = A_p - A_r = \frac{\pi(D^2 - d^2)}{4} = \frac{\pi(10^2 - 6^2)}{4} = 50.27 \text{ cm}^2$$

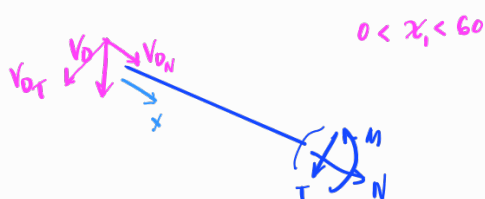
$$J_x = \frac{\pi(D^4 - d^4)}{64} = 427.26 \text{ mm}^4$$

- calcolare lo spostamento orizzontale del punto A,



$$\begin{cases} R_{x=0} & H_B + 1 = 0 & \boxed{H_B = -1 \text{ N}} \\ R_{y=0} & V_B - V_D = 0 & \boxed{V_B = V_D = 0.54 \text{ N}} \\ M_{B=0} & -V_D \cdot (150 + 50) + 1 \cdot (50 \cdot \tan \beta) = 0 & \boxed{V_D = 0.536 \text{ N}} \end{cases}$$

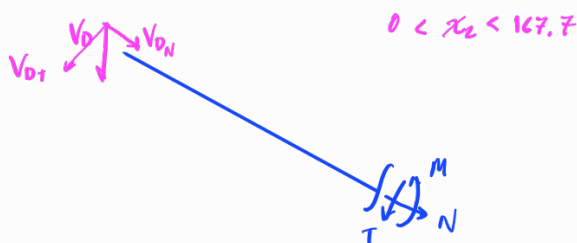
- tracciare i diagrammi delle azioni interne;



$$N = -V_{DN} = -V_D \cdot \sin \alpha = -0.24 \text{ N}$$

$$T = -V_{DT} = -V_D \cdot \cos \alpha = -0.48 \text{ N}$$

$$M = -0.148 x_1$$



$$N = -V_{DN} = -V_D \cdot \sin \alpha = -0.24 \text{ N}$$

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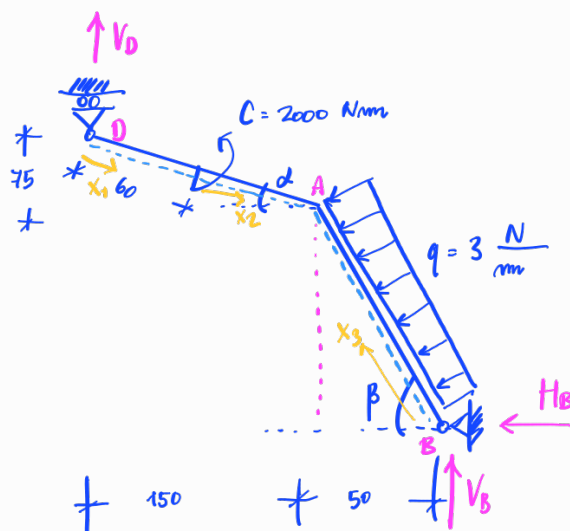
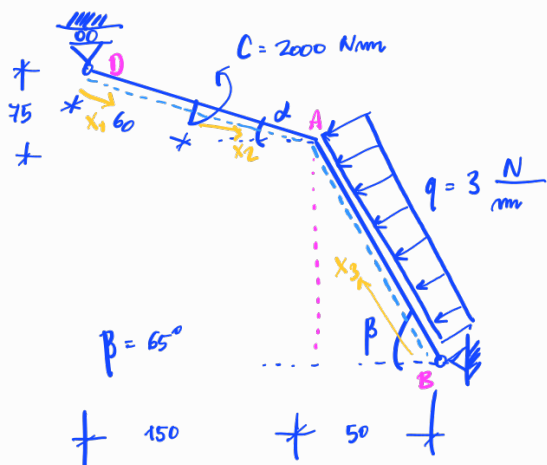


$$N = -V_{B,N} \cdot \sin \beta - H_{B,N} \cdot \cos \beta = -0,91 \text{ N}$$

$$T = -V_{B,T} \cdot \cos \beta + H_{B,T} \cdot \sin \beta = 0,68 \text{ N}$$

$$M = (+V_{B,T} \cdot \cos \beta - H_{B,T} \cdot \sin \beta) \cdot x_3 = -0,68 \cdot x_3$$

- Spostamenti



$$\left\{ \begin{array}{l} R_H = 0 \\ R_V = 0 \\ M_B = 0 \end{array} \right.$$